AI AVATAR

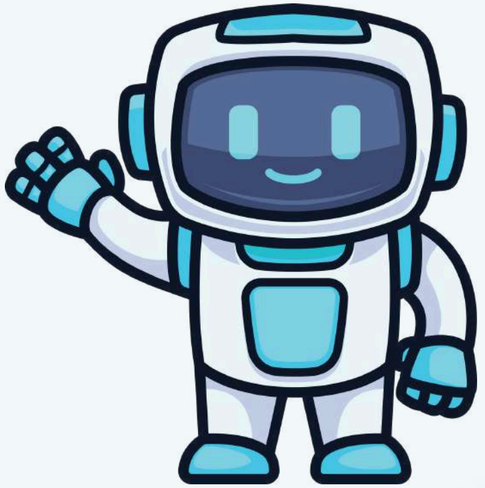
CHATBOT

An Intelligent Conversational Assistant for Students & Visitors

TEAM MEMBERS

Branch: CSE(AI & ML)

Krishna Kushwah
Om channawar
Manmath Kornule
Arjun Mankuskar
Pratik Mane



INTRODUCTION

Revolutionizing Student Assistance Through AI

- **In today’s fast-paced world, students, parents, and visitors seek instant information about college facilities, placements, fees, and more.**

- **Our AI Avatar Chatbot provides an interactive, intelligent, and user-friendly way to resolve queries 24/7.**

PROBLEM STATEMENT

THE CHALLENGE WE'RE SOLVING

Every day, students, parents, and visitors have hundreds of questions about our college:

- "What's the latest placement report?"
- "Where can I check the hostel facilities?"
- "How does the NEP pattern affect us?"



But the current process of finding answers is manual, time-consuming, and scattered across multiple sources.
There's no single platform that provides instant, reliable, and 24/7 assistance.

PROJECT VISION

OUR VISION

“To create an intelligent, interactive, and scalable AI-powered avatar chatbot that delivers instant, accurate, and personalized assistance to students, parents, and visitors — making college communication smarter and seamless.”

Highlights:

- Bridging the gap between students and administration
- Offering 24/7 query resolution
- Continuously updatable knowledge base
- A modern AI solution for next-gen campuses

KEY FEATURES

1. Dual Input Modes
Students can ask queries through text or voice, making interaction flexible and accessible.

2. Realistic Voice Interaction
Chatbot replies are converted into natural speech using Text-to-Speech (TTS).

3. Knowledge Base Integration
Institution-specific documents (PDFs, policies, curriculum) are connected for accurate answers.

4. Real-Time Response Streaming
Replies appear instantly in the chat, giving a live conversation experience.

5. 3D Avatar
3D avatar with expressions make interaction more engaging and human-like.

TECHNOLOGIES USED

- **Frontend:** HTML, CSS, JS (chat UI, floating input, streaming chat)
- **Backend:** Flask, FastAPI (routes, API, database handling)
- **AI/ML:** Ollama models for response generation
- **Voice:** ElevenLabs TTS, Speech-to-Text integration
- **Docker:** Deployment

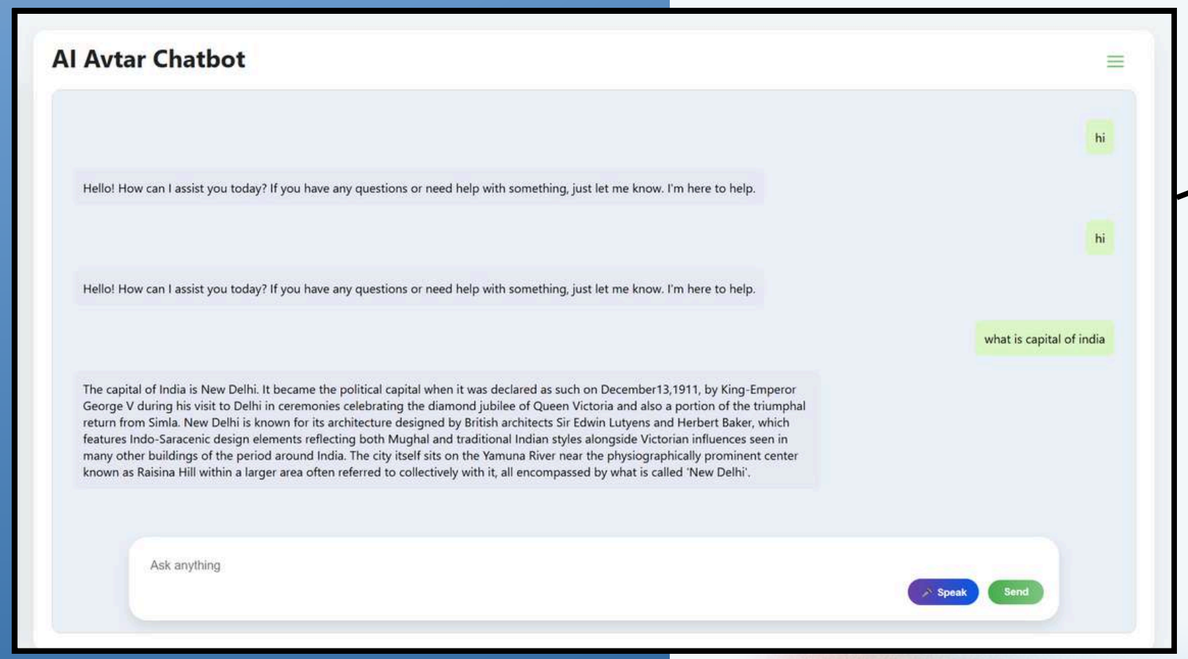


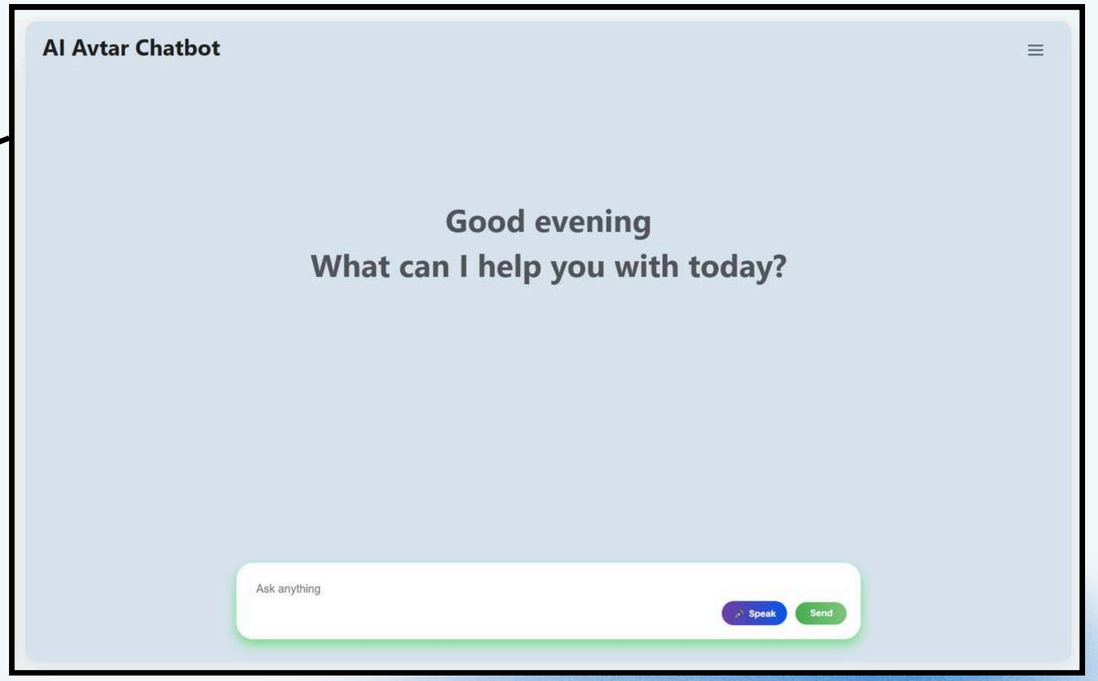
TEAM CONTRIBUTIONS

MEMBER 1: MANMATH KORNULE

Title: Frontend & Chat UI

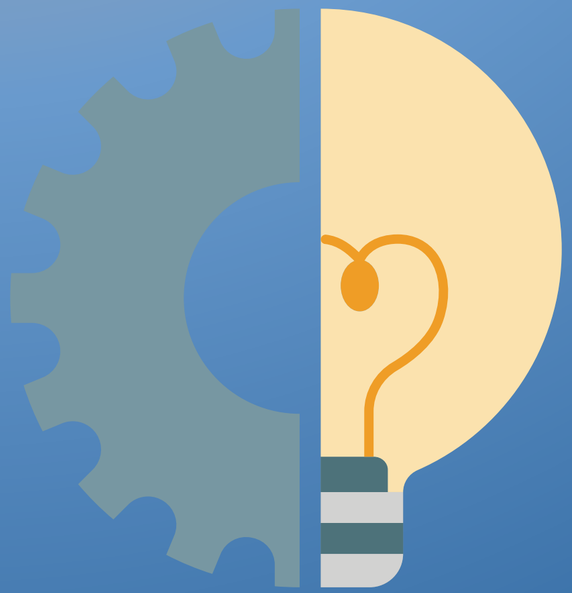
- Designed User Interface of the chatbot using **Flask ,html ,css, js**
- Implemented **real-time response streaming to reduce response time.**
- Integrated **“Speak” button** for TTS activation
- integrated **fasterwhisper** model for STT(speech to text)
- **Connected** and **Tested** frontend with Flask backend routes.
- used **SQLite** database to save metadata of documents





TEAM CONTRIBUTIONS

MEMBER 2 : KRISHNA KUSHWAH



Backend and Response

-A Multi-Layered Approach for Speed, Accuracy, and Intelligence

Core Technologies (The Stack) ⚙️

Logo	Technology	Role
	FastAPI	High-performance, asynchronous API framework to handle all user requests.
	Ollama	Runs local Large Language Models (e.g., Mistral) for intelligent text generation and intent classification.
	LangChain	Handles document loading and splitting during the data ingestion phase.
	ChromaDB	Vector database for storing document embeddings and enabling fast semantic search.
	SQLite	Lightweight database for the high-speed Knowledge Base (precise FAQs).

TEAM CONTRIBUTIONS

MEMBER 2 : KRISHNA KUSHWAH



Backend and Response

🌊 Intent-Driven Query Processing Workflow

1. Query & Intent Classification

- FastAPI receives query.
- LLM classifies intent:
  - Informational (MITAOE-related)
  - Small talk (hello, thanks, etc.)

2. Path A: Informational Query (RAG Pipeline)

- Layer 1 (Cache): Check TTLCache (<0.1s).
- Layer 2 (KB): SQLite DB semantic match (>95%).
- Layer 3 (RAG): Retrieve from ChromaDB → Inject KB examples → Generate via Ollama.

3. Path B: Small Talk Query

- Skip pipeline → LLM generates quick friendly reply.

4. Final Output (Stream & Cache)

- Stream tokens live to frontend.
- Smart post-processing (fix spacing, contractions).
- Cleaned response cached for reuse.

TEAM CONTRIBUTIONS

MEMBER 3:
OM CHANNAWAR



Title: Voice Input & Transcription

- Implemented **speech-to-text (STT)** using **Faster Whisper** for fast, accurate transcription.
- Captures user voice through the browser microphone using MediaRecorder.
- Sends **transcribed text to the chatbot backend** to generate responses instantly.
- Worked on **3D avtar**



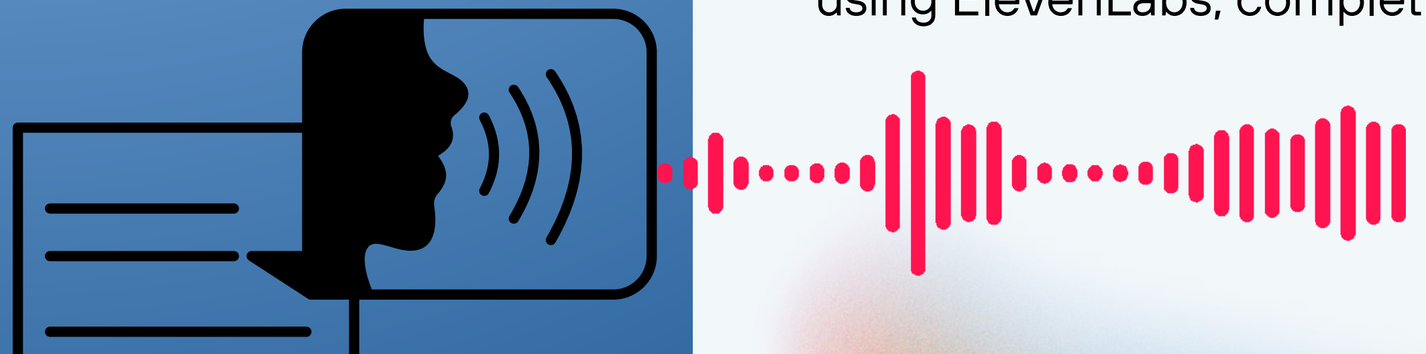
TEAM CONTRIBUTIONS

MEMBER 4:
ARJUN MANUSKAR

Title: Voice Generation Module

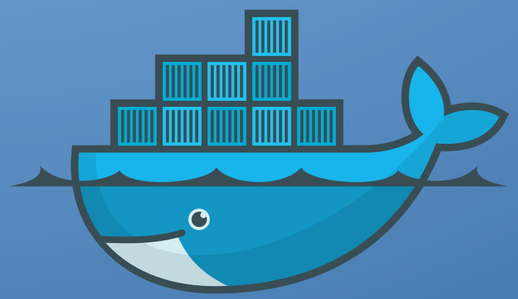
- Tasks Completed:
- Set up secure **API access using environment variables**
- Retrieved and tested available **voice profiles**
- Selected a suitable voice (George) for **output consistency**
- Integrated a **dynamic input pipeline** where the output from a **language model (LLM)** is captured and passed into the voice generation module. The received text is automatically converted into an **.mp3** file using **ElevenLabs API.**

- **Outcome:** Successfully converted text into high-quality speech output using ElevenLabs, completing the voice synthesis module.



TEAM CONTRIBUTIONS

MEMBER 5: PRATIK MANE

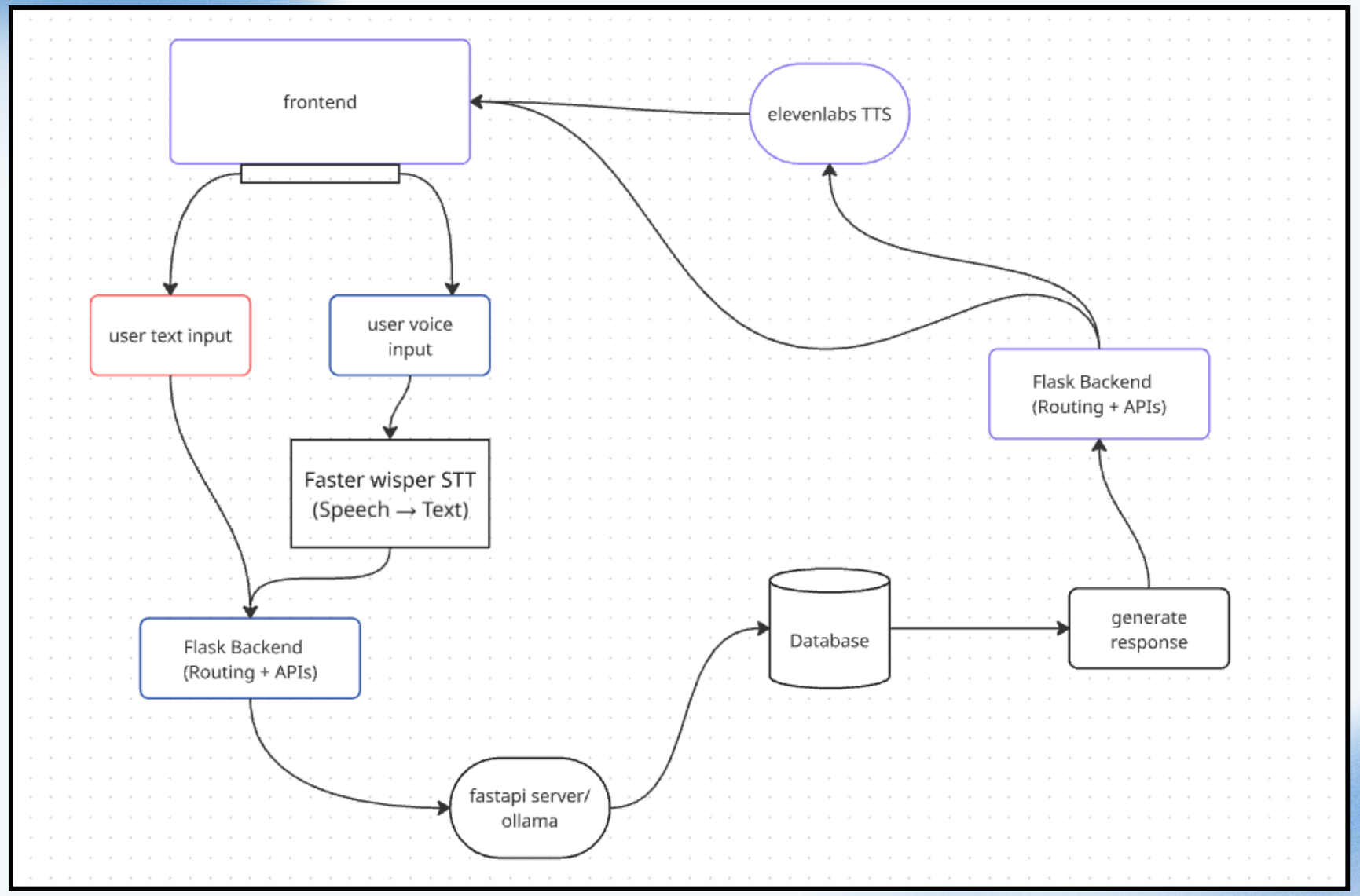


Title: Docker Integration

DEPLOYMENT & SYSTEM INTEGRATION

- Created **Dockerfiles for both Backend (FastAPI) and Frontend (Flask).**
- Ensured portability by **containerizing all services with dependencies.**
- **Exposed required ports** (8000 for backend, 5000 for frontend).
- Managed **static files, uploads, and audio directories in containers.**
- Enabled smooth **API communication between containers.**
- Prepared system for **future scalability** (Docker Compose / Kubernetes).

SYSTEM WORKFLOW



LIVE DEMO

THANK YOU

